

IMO Training

Sequences and Introduction to Analysis

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Introduction

Definition. (Convergence of Sequence) Let (a_n) be a sequence and $\ell \in \mathbb{R}$. Then a_n *converges to ℓ , tends to ℓ , or $a_n \rightarrow \ell$* , if for all $\varepsilon > 0$, there is some $n \in \mathbb{N}$ such that whenever $n > N$, we have $|a_n - \ell| < \varepsilon$. In symbols, this says

$$(\forall \varepsilon > 0)(\exists N)(\forall n \geq N) |a_n - \ell| < \varepsilon.$$

We say ℓ is the *limit* of (a_n) .

Definition. (Continuous Function) Let $A \subseteq \mathbb{R}$, $a \in A$, and $f : A \rightarrow \mathbb{R}$. Then f is *continuous at a* if for any $\varepsilon > 0$, there is some $\delta > 0$ such that if $y \in A$ is such that $|y - a| < \delta$, then $|f(y) - f(a)| < \varepsilon$. In symbols, we have

$$(\forall \varepsilon > 0)(\exists \delta > 0)(\forall y \in A) |y - a| < \delta \Rightarrow |f(y) - f(a)| < \varepsilon.$$

f is *continuous* if it is continuous at every $a \in A$. In symbols, we have

$$(\forall a \in A)(\forall \varepsilon > 0)(\exists \delta > 0)(\forall y \in A) |y - a| < \delta \Rightarrow |f(y) - f(a)| < \varepsilon.$$

Theorem. (Bolzano-Weierstrass) Every bounded sequence has a convergent subsequence.

Lemma. (Comparison Test) Let (a_n) and (b_n) be non-negative sequences, and suppose that $\exists C, N$ such that $\forall n \geq N$, $a_n \leq Cb_n$. Then if $\sum b_n$ converges, then so does $\sum a_n$.

Lemma. (Alternating Series Test) Let (a_n) be a decreasing sequence of non-negative reals, and suppose that $a_n \rightarrow 0$. Then $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges, i.e. $a_1 - a_2 + a_3 - a_4 + \cdots$ converges.

Lemma. (Integral Test) Let $f : [1, \infty] \rightarrow \mathbb{R}$ be a decreasing non-negative function. Then $\sum_{n=1}^{\infty} f(n)$ converges iff $\int_1^{\infty} f(x) dx < \infty$.

Lemma. (Ratio Test)

1. If $\exists c < 1$ such that $\frac{|a_{n+1}|}{|a_n|} \leq c$ for all n , then $\sum a_n$ converges.
2. If $\exists c < 1$ and $\exists N$ such that $(\forall n \geq N) \frac{|a_{n+1}|}{|a_n|} \leq c$, then $\sum a_n$ converges. Note that just because the ratio is always less than 1, it doesn't necessarily converge. It has to be always less than a fixed number c . Otherwise the test will say that $\sum 1/n$ converges.
3. If $\exists \rho \in (-1, 1)$ such that $\frac{a_{n+1}}{a_n} \rightarrow \rho$, then $\sum a_n$ converges. Note that we have the *open* interval $(-1, 1)$. If $\frac{|a_{n+1}|}{|a_n|} \rightarrow 1$, then the test is inconclusive!

Theorem. (Stolz–Cesàro) Let $(a_n)_{n \geq 1}, (b_n)_{n \geq 1}$ be two real sequences. It is given that either one of the following holds:

1. (b_n) is a strictly monotone and divergent sequence; or
2. $a_n \rightarrow 0$ and $b_n \rightarrow 0$, and (b_n) is strictly decreasing,

then $\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = L$ implies $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$.

Problems

1. (Vietnam 2021 1) Define $(x_n)_{n \geq 1}$ by $0 < x_1 < \frac{1}{2}$ and for all $n \geq 1$, $x_{n+1} = 3x_n^2 - 2nx_n^3$.
 - (a) Prove that $x_n \rightarrow 0$.
 - (b) For each $n \geq 1$, let $y_n = \sum_{k=1}^n kx_k$. Prove that (y_n) converges.
2. (Vietnam 2020 1) Define sequence $(x_n)_{n \geq 1}$ by $x_1 = 1$, and for all $n \geq 1$, $x_{n+1} = x_n + 3\sqrt{x_n} + \frac{n}{\sqrt{x_n}}$.
 - (a) Prove that $\frac{n}{x_n} \rightarrow 0$.
 - (b) Find $\lim_{n \rightarrow \infty} \frac{n^2}{x_n}$.
3. (Vietnam 2019 1) Let $f : \mathbb{R} \rightarrow (0, +\infty)$ be a continuous function such that $\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow +\infty} f(x) = 0$.
 - (a) Prove that $f(x)$ has the maximum value on $\mathbb{R}$.
 - (b) Prove that there exists two sequences $(x_n), (y_n)$ such that for all $n \geq 1$, $x_n < y_n, f(x_n) = f(y_n)$, and $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} y_n$.

4. (Vietnam 2018 1) Define sequence $(x_n)_{n \geq 1}$ by $x_1 = 2$, and for all $n \geq 1$,
 $x_{n+1} = \sqrt{x_n + 8} - \sqrt{x_n + 3}$.
- (a) Prove that (x_n) converges, and find the limit.
- (b) Prove that for all $n \geq 1$, $n \leq \sum_{k=1}^n x_k \leq n + 1$.
5. (Vietnam 2017 1) For $a \in \mathbb{R}$, define sequence $(u_n)_{n \geq 1}$ by $u_1 = a$, and for all $n \geq 1$,
 $u_{n+1} = \frac{1}{2} + \sqrt{\frac{2n+3}{n+1}u_n + \frac{1}{4}}$.
- (a) Prove that (u_n) converges when $a = 5$, and find the corresponding limit.
- (b) Find all a such that (u_n) is well-defined and converges.
6. (Vietnam 2015 1) For non-negative $a \in \mathbb{R}$, define sequence $(u_n)_{n \geq 1}$ by $u_1 = 2$, and for all
 $n \geq 1$, $u_{n+1} = \frac{u_n}{2} + \frac{n^2}{4n^2 + a} \sqrt{u_n^2 + 3}$.
- (a) Prove that (u_n) converges when $a = 0$, and find its limit.
- (b) Prove that (u_n) converges when $a \in [0, 1]$.
7. (Vietnam 2014 1) Define sequences $(x_n)_{n \geq 1}, (y_n)_{n \geq 1}$ by $x_1 = 1, y_1 = \sqrt{3}$, and for all $n \geq 1$,

$$\begin{cases} x_{n+1}y_{n+1} - x_n &= 0 \\ x_{n+1}^2 + y_n &= 2 \end{cases}$$

Prove that the sequences converge and find their limits.

8. (Vietnam 2013 2) Define sequence $(a_n)_{n \geq 1}$ by $a_1 = 1$, and for all $n \geq 1$, $a_{n+1} = 3 - \frac{a_n + 2}{2^{a_n}}$.
 Prove that (a_n) converges, and find its limit.
9. (Vietnam 2012 1) Define sequence $(x_n)_{n \geq 1}$ by $x_1 = 3$, and for all $n \geq 2$,
 $x_n = \frac{n+2}{3n}(x_{n-1} + 2)$. Prove that (x_n) converges, and find its limit.
10. (Vietnam 2011 2) Define real sequence (x_n) by $x_1 = 1$ and for all $n \geq 2$,
 $x_n = \frac{2n}{(n-1)^2} \sum_{i=1}^{n-1} x_i$. Prove that $\lim_{n \rightarrow \infty} x_{n+1} - x_n$ exists and is finite.
11. (Putnam 1988 A3) Determine, with proof, the set of real numbers x for which
 $\sum_{n=1}^{\infty} \left(\frac{1}{n} \csc \frac{1}{n} - 1 \right)^x$ converges.
12. (Putnam 1988 B4) Prove that if $\sum_{n=1}^{\infty} a_n$ is a convergent series of positive real numbers,
 then so is $\sum_{n=1}^{\infty} a_n^{\frac{n}{n+1}}$.
13. (Putnam 1990 A2) Is $\sqrt{2}$ the limit of a sequence of numbers of the form $\sqrt[n]{n} - \sqrt[n]{m}$, where
 $n, m \in \mathbb{Z}_{\geq 0}$?

14. (Putnam 2008 A4) Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} x & \text{if } x \leq e \\ xf(\ln x) & \text{if } x > e \end{cases}$$

Does $\sum_{n=1}^{\infty} \frac{1}{f(n)}$ converge?

15. (Putnam 2016 B1) Define sequence $(a_n)_{n \geq 0}$ by $x_0 = 1$ and for $n \geq 0$, $x_{n+1} = \ln(e^{x_n} - x_n)$.

Show that $\sum_{n=0}^{\infty} x_n$ converges, and find its sum.

16. (Putnam 2019 B2) For all $n \geq 2$, let

$$a_n = \sum_{k=1}^{n-1} \frac{\sin\left(\frac{(2k-1)\pi}{2n}\right)}{\cos^2\left(\frac{(k-1)\pi}{2n}\right) \cos^2\left(\frac{k\pi}{2n}\right)}.$$

Determine $\lim_{n \rightarrow \infty} \frac{a_n}{n^3}$.

17. (Putnam 2020 A3) Let $a_0 = \frac{\pi}{2}$, and let $a_n = \sin(a_{n-1})$ for $n \geq 1$. Determine whether

$\sum_{n=1}^{\infty} a_n^2$ converges.

18. (Putnam 2021 A2) For every $x \in \mathbb{R}^+$, let $g(x) = \lim_{r \rightarrow 0} \left((x+1)^{r+1} - x^{r+1}\right)^{\frac{1}{r}}$.

Find $\lim_{x \rightarrow \infty} \frac{g(x)}{x}$.